Part IV

BGM Built From the Ground Up

CHAPTER 12

Change of Numeraire and the Forward Measure

12.1 The one idea this whole book leans on

Every pricing formula used so far — Black, Bachelier, for a caplet or a swaption — was justified by a claim asserted without proof: "under *this* measure, *this* rate is a driftless martingale." This chapter proves that claim, in general, and in doing so hands you the single tool that makes the rest of the book possible: the ability to move freely between different ways of measuring value, choosing whichever one makes a given problem driftless and therefore tractable.

Key result: the numeraire and the pricing measure

A **numeraire** is any strictly positive, tradable asset $N(t)$. The fundamental theorem of asset pricing states that, for a market free of arbitrage, there exists a measure $\mathbb{Q}^N$ (the **measure associated with numeraire** N) under which the price of *every* other tradable asset $V(t)$, when expressed in units of N, is a martingale:

$$\frac{V(t)}{N(t)} = \mathbb{E}_t^{\mathbb{Q}^N}\left[\frac{V(T)}{N(T)}\right] \qquad \Longrightarrow \qquad V(t) = N(t)\,\mathbb{E}_t^{\mathbb{Q}^N}\left[\frac{V(T)}{N(T)}\right]$$

Different choices of N give different measures $\mathbb{Q}^N$, all consistent, all giving the same price $V(t)$ — but some choices make a *specific* $V(T)/N(T)$ dramatically simpler to compute than others. Choosing the numeraire well is most of the art of pricing a rates derivative in closed or semi-closed form.

12.2 Two numeraires already used, made explicit

Chapters 6 and 7 each made an implicit numeraire choice; naming them now makes the pattern explicit and reusable.

Key result: the forward measure and the annuity measure

Choosing $N(t) = P(t, T_2)$ (a zero-coupon bond maturing at the caplet's payment date) gives the **T_2-forward measure** $\mathbb{Q}^{T_2}$. Under this measure, the simple forward rate $L(t; T_1, T_2)$ — itself expressible as a portfolio of two zero-coupon bonds divided by $P(t, T_2)$, hence itself a traded/numeraire-relative price — is automatically a driftless martingale. This is exactly why Chapter 6 could assert, without further justification, that F_t has zero drift.
Choosing $N(t) = A(t)$ (the swap annuity from Chapter 4) gives the **annuity measure** $\mathbb{Q}^A$ (also called the **swap measure**). Under this measure, the forward swap rate $S(t)$ — itself expressible, from Chapter 4, as $(P(t, T_0) - P(t, T_n))/A(t)$, again a numeraire-relative traded price — is automatically a driftless martingale, justifying Chapter 7's swaption formula in exactly the same way.

12.3 Girsanov's theorem: how drift changes under a change of measure

The mechanical engine behind all of this is Girsanov's theorem, which describes precisely how a Brownian motion's drift transforms when the measure changes.

Key result: Girsanov's theorem, pricing form

If $W(t)$ is a Brownian motion under measure $\mathbb{Q}$, and $\mathbb{Q}'$ is an equivalent measure defined via a change-of-numeraire ratio whose volatility is $\gamma(t)$, then

$$W'(t) = W(t) - \int_0^t \gamma(s)\,\mathrm{d}s$$

is a Brownian motion under $\mathbb{Q}'$. Equivalently: switching from $\mathbb{Q}$ to $\mathbb{Q}'$ adds a drift of $\gamma(t)\,\mathrm{d}t$ to every process driven by $W(t)$, where $\gamma(t)$ is the volatility of the ratio of the new numeraire to the old one. The *diffusion* (volatility) coefficient of any process is completely unaffected by a change of numeraire — only the drift changes. This single fact is what makes change-of-numeraire tractable at all: you never have to redo the hard part (specifying volatility), only recompute a drift correction.

Worked example 12.1 — the mechanics of a drift correction, in miniature.
Suppose, under measure $\mathbb{Q}$, a process $X(t)$ has dynamics $\mathrm{d}X = \sigma X\,\mathrm{d}W(t)$ (driftless), and the ratio of a new numeraire to the old one, $\rho(t) = N'(t)/N(t)$, has its own lognormal volatility γ, i.e. $\mathrm{d}\rho/\rho = (\dots)\mathrm{d}t + \gamma\,\mathrm{d}W(t)$, sharing the same driving Brownian motion as X with instantaneous correlation 1 for this illustration.
Step 1 — apply Girsanov. Under the new measure $\mathbb{Q}'$, $W'(t) = W(t) - \int_0^t \gamma\,\mathrm{d}s$ is Brownian, so $W(t) = W'(t) + \int_0^t \gamma\,\mathrm{d}s$.
Step 2 — substitute into X's dynamics.

$$\mathrm{d}X = \sigma X\,\mathrm{d}W = \sigma X\left(\mathrm{d}W' + \gamma\,\mathrm{d}t\right) = \sigma\gamma X\,\mathrm{d}t + \sigma X\,\mathrm{d}W'$$

Step 3 — read off the result. Under $\mathbb{Q}'$, X now has drift $\sigma\gamma X$ — proportional to X's own volatility σ *times* the new numeraire's volatility γ. The diffusion coefficient, σX, is completely unchanged.
Interpretation. This tiny calculation is, in essence, the entire BGM drift formula of Chapter 13 in miniature: a forward rate's drift under some measure other than its own is always its own volatility multiplied by the volatility of whatever numeraire ratio connects its natural measure to the working measure, summed appropriately when multiple rates and correlations are involved. Chapter 13 does nothing conceptually new beyond this — it just carries out this same substitution for the specific numeraire ratio relevant to a full BGM curve.

PITFALL

It is easy to mentally conflate "driftless under some measure" with "driftless, full stop." A quantity is only ever driftless *relative to a specific numeraire*. A forward rate is driftless under its own forward measure and has a nonzero drift under literally every other measure — including, crucially, the measures needed to evolve a whole curve of forward rates simultaneously, which is exactly the situation Chapter 13 confronts. Reading a result like "$L_i(t)$ is a martingale" without immediately asking "under which measure?" is a reliable way to misapply it.

ON THE DESK

Change of numeraire is not merely a theoretical nicety; it is the reason a quant library organises its pricing code around "which measure is this computation being done under," often as an explicit parameter or context object passed through the simulation engine. A single BGM Monte Carlo run typically fixes *one* common measure (Chapter 13 discusses the two standard choices — spot and terminal) for the whole simulation, and every individual forward rate's simulated drift is computed relative to that one fixed choice, even though each rate would individually prefer a different, simpler measure of its own. Getting this bookkeeping right, consistently, across a large codebase handling many products, is a genuine and recurring source of implementation bugs.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. State the fundamental pricing identity relating a traded asset's price to a numeraire and the associated pricing measure.
2. Explain what the forward measure and annuity measure are, and why each makes its associated rate (L or S) driftless.
3. State Girsanov's theorem in the form used for pricing, and explain why a change of measure alters drift but never volatility.